

Relation Algebra

§14. Thinning Algorithms

§14.1. Thinning

(14.1a)

For $Q : A \rightarrow A$, $\text{thin}(Q) \triangleq (\exists/\exists) \cap (\epsilon \setminus (Q^\circ \epsilon)) : EA \rightarrow EA$ (8.1).

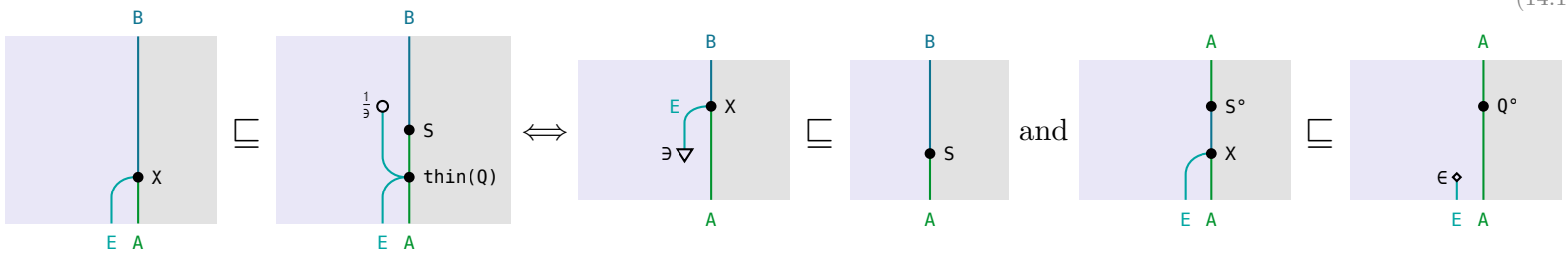
$ys \text{ thin}(Q) xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \ Q \ a)$ the same $^\circ$ as $\triangleq \text{est}$: ϵX runs $a \rightarrow ys \rightarrow xs$, member of ys first, and $Q^\circ \epsilon$ runs $a \rightarrow b \rightarrow xs$, so $(a, b) \in Q^\circ$ reads $b \ Q \ a$; at $Q \triangleq \leq$ every $a \in ys$ keeps some $b \leq a$ in xs , the end $\text{est}(\leq)$ picks

(14.1b)

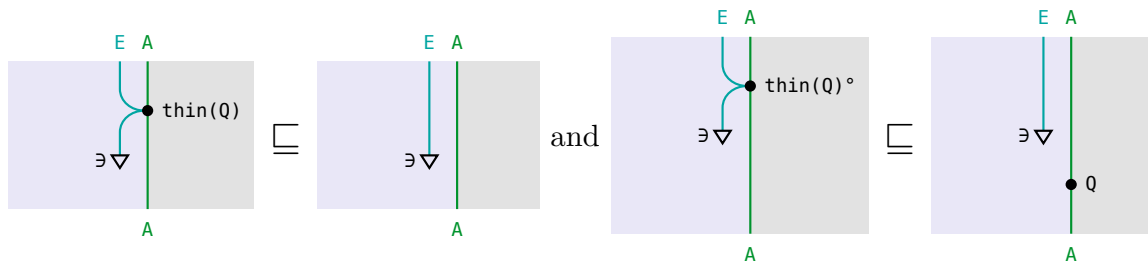
the law	what it says
$X \subseteq \frac{S}{\exists} \text{thin}(Q) \iff X \exists \in S$ and $S^\circ X \in Q^\circ \epsilon$	everything kept is an S-value, and every S-value has a Q-lower bound among the kept ones
$\text{thin}(Q) \exists \in \exists$	everything a thinning keeps was in the set
$\text{thin}(Q)^\circ \exists \in \exists Q$	every element of the set has a Q-lower bound among the kept ones
$Q \subseteq R \implies \text{thin}(Q) \subseteq \text{thin}(R)$	the fewer pairs Q relates, the fewer subsets count as thinnings
$1 \subseteq Q \implies 1 \subseteq \text{thin}(Q)$ $Q \subseteq Q \implies \text{thin}(Q) \text{thin}(Q) \subseteq \text{thin}(Q)$	keeping everything is always a legal thinning
$Q \subseteq R \wedge 1 \subseteq Q \wedge R R \subseteq R \implies \text{est}(R) = \text{thin}(Q) \text{est}(R)$ $Q \subseteq R, 1 \subseteq Q, R R \subseteq R$ — (14.1.2a); weaker than the book's "both preorders": Q transitive is never used and $1 \subseteq R$ follows	thin-introduction : thinning first cannot lose an R-minimum
$\text{thin}(Q) \supseteq \text{est}(Q) \frac{1}{\exists}$ (8.2) — (14.1.2d)	thin-elimination : keeping one element is a thinning, but its domain is the sets $\text{est}(Q)$ is defined on
$\frac{S}{\exists} \text{thin}(Q) \supseteq \frac{S}{\exists} \text{est}(R) \frac{1}{\exists}$ (8.3), $R \cap (S^\circ S) \subseteq Q$ — thin-elimination variant	the usable variant: R need only refine Q between values S gives one argument
$P(\text{thin}(Q)) \text{union} \subseteq \text{union thin}(Q)$ (8.4)	thinning each member set is a thinning of the union

§14.1.1. $X \in (S^\circ \exists) \text{thin}(Q) \iff X \exists \in S$ and $S^\circ X \in Q^\circ \epsilon$

(14.1.1a)



(14.1.1b)



(14.1.1a) at $X \triangleq \text{thin}(Q)$, $S \triangleq \exists$: the left side is $\text{thin}(Q) \subseteq \text{thin}(Q)$, since $\exists \exists = \wedge(\exists) = 1$

§14.1.2. $\text{est}(R) = \text{thin}(Q) \text{est}(R)$ given $Q \subseteq R$, $1 \subseteq Q$, $R R \subseteq R$

(14.1.2a)

$\text{est}(R) \sqsubseteq \text{thin}(Q) \text{ est}(R)$

the $\sqsubseteq$ half: keeping everything is a thinning — $1 \sqsubseteq Q$

(a) $1 \sqsubseteq \text{thin}(Q)$ — (14.1b)

(14.1.2b)

$\text{thin}(Q) \text{ est}(R) \sqsubseteq \exists$

the $\sqsubseteq$ half, first condition of the UP of est at $X \triangleq \text{thin}(Q) \text{ est}(R)$ — est-up

(a) $\text{est}(R) \sqsubseteq \exists$ — est-laws

(b) $\text{thin}(Q) \sqsubseteq \exists$ — (14.1.1a)

(14.1.2c)

$\in \text{thin}(Q) \text{ est}(R) \sqsubseteq R^\circ$

the $\sqsubseteq$ half, second condition — $Q \sqsubseteq R$, R transitive

(a) $\in \text{thin}(Q) \sqsubseteq Q^\circ \in$ — (14.1.1a)

(b) $\in \text{est}(R) \sqsubseteq R^\circ$ — est-laws

(14.1.2d)

$\text{est}(Q) \frac{1}{\exists} \sqsubseteq \text{thin}(Q)$

the singleton holding a Q -least member of a set is a thinning of that set

$\text{est}(Q) \frac{1}{\exists} \sqsubseteq \exists$

every member of the singleton is a member of the set

(a) $\frac{1}{\exists} \sqsubseteq 1$ — pow-laws ; $\text{est}(Q) \sqsubseteq \exists$ — est-laws

$\in \text{est}(Q) \frac{1}{\exists} \sqsubseteq Q^\circ \in$

the singleton's member stands in Q to every member of the set

(a) $\triangleq \text{thin}, \exists \backslash (Q^\circ \in)$ half

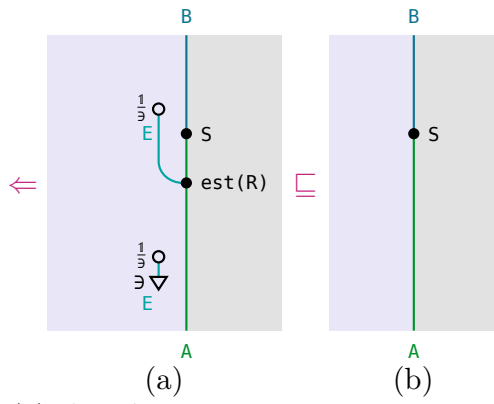
(b) $\in \text{est}(Q) \sqsubseteq Q^\circ$ — est-up

$$\text{RnS}^\circ \text{S} \sqsubseteq \text{Q} \Rightarrow \frac{\text{S}}{\exists} \text{est}(\text{R}) \frac{1}{\exists} \sqsubseteq \frac{\text{S}}{\exists} \text{thin}(\text{Q})$$

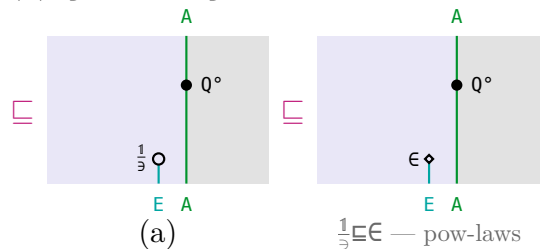
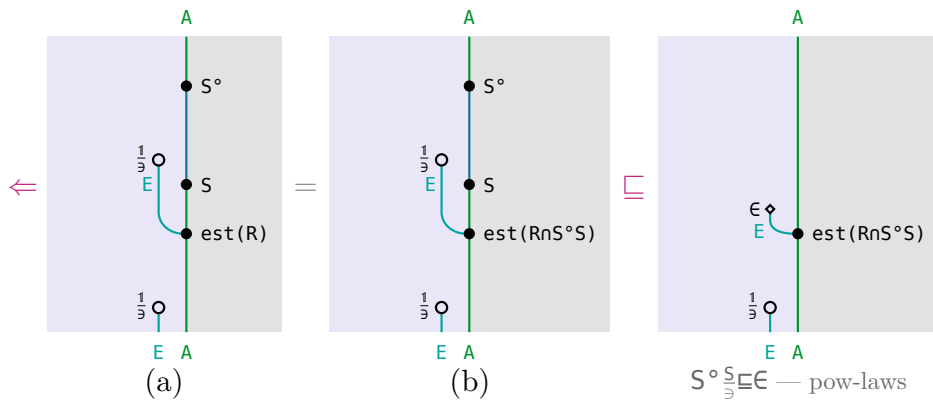
given $\text{Rn}(\text{S}^\circ \text{S}) \sqsubseteq \text{Q}$, Q a preorder

(14.1.2e)

$$\frac{\text{S}}{\exists} \text{est}(\text{R}) \frac{1}{\exists} \sqsubseteq \text{S}$$

 every member of the singleton is a value S returns


$$\text{RnS}^\circ \text{S} \sqsubseteq \text{Q} \Rightarrow \text{S}^\circ \frac{\text{S}}{\exists} \text{est}(\text{R}) \frac{1}{\exists} \sqsubseteq \text{Q}^\circ \sqsubseteq$$

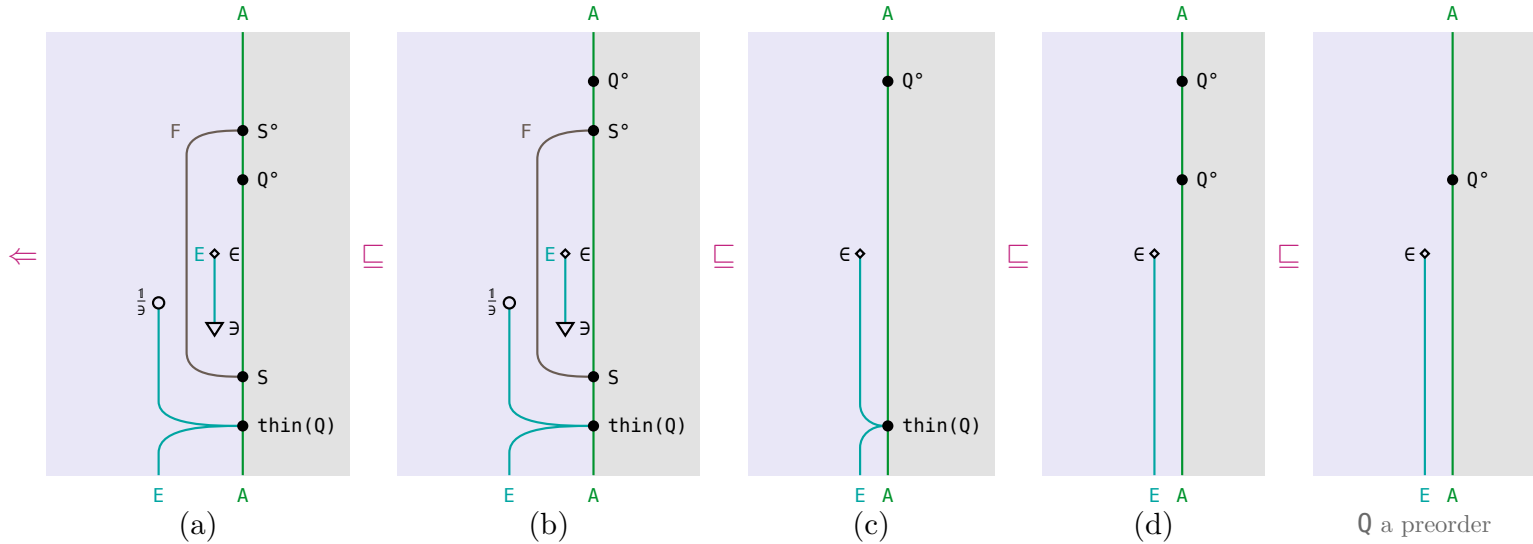
 the singleton's member stands in Q to every value S returns at the same argument


$$\text{PreservesRecip}(F) \wedge Q \sqsubseteq Q \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \sqsubseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle \quad (14.1.2f)$$

thinning at every step of the reduce is a thinning of the whole candidate set — S monotonic on Q , Q a preorder

$$\langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \sqsubseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle \quad \text{and} \quad \langle \frac{(S)}{\exists} \rangle \circ \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \sqsubseteq Q^\circ \in$$

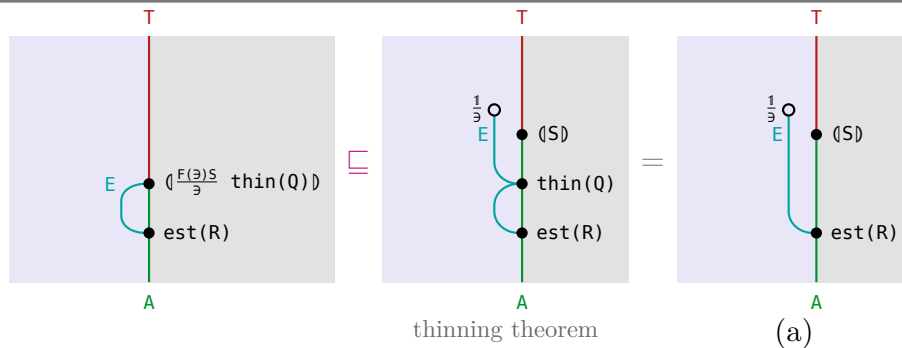
(14.1b) at $X \triangleq \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle$, $\langle \frac{(S)}{\exists} \rangle$ for its S



- (a) fusion, hylomorphism theorem
 (b) $S^\circ F(Q^\circ) \sqsubseteq Q^\circ S^\circ$ — mon-str, relator-laws
 (c) $S^\circ F(\epsilon) \langle \frac{F(\exists)S}{\exists} \rangle \sqsubseteq \epsilon$ — pow-laws
 (d) $\epsilon \text{thin}(Q) \sqsubseteq Q^\circ \epsilon$ — $\triangleq$ thin, adj-all

$$\text{PreservesRecip}(F) \wedge Q \in R \wedge 1 \in Q \wedge Q \sqsubseteq Q \wedge R \in R \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \text{est}(R) \sqsubseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle \text{est}(R) \quad (14.1.2g)$$

the thinning fold refines the optimisation problem itself — S monotonic on Q , $Q \in R$, both preorders



- (a) $\text{est}(R) = \text{thin}(Q) \text{est}(R)$ — (14.1b)

§14.2. Paths in a layered network

(14.2a)

$F(A, X) = A + A \times X$, $L = \text{list}^+$ with initial algebra $\alpha \triangleq [\text{wrap}, \text{cons}] : F(A, LA) \rightarrow LA$.

$\text{wrapz} \triangleq \langle \text{wrap}, \text{zero} \rangle$, $\text{consw}(a, (xs, n)) = (\text{cons}(a, xs), \text{wt}(a, \text{head}(xs)) + n)$.

$\text{cost} \triangleq \langle [\text{wrapz}, \text{consw}] \rangle \pi_2$, $\langle [\text{wrapz}, \text{consw}] \rangle = \langle 1, \text{cost} \rangle$, $R \triangleq \text{cost} \leq \text{cost}^\circ$.

$Q \triangleq R \circ (\text{head} \text{head}^\circ)$, $S \triangleq F(1, \exists) \alpha$, $\frac{F(\exists, 1)}{\exists} = 1 + \text{cpl}$, $\frac{F(1, \exists)}{\exists} = 1 + \text{cpr}$, $\text{step} \triangleq \text{cpr } P(\text{cons}) \text{est}(R)$.

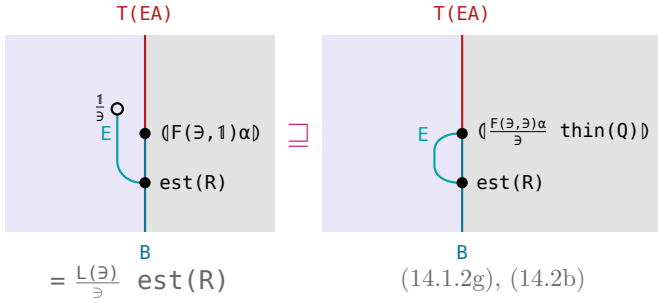
(14.2b)

the law	what it says
$F(\exists, Q) \alpha \sqsubseteq F(\exists, 1) \alpha Q$	$F(\exists, 1) \alpha$ is monotonic on Q ; on R it is not, since the next edge can cost arbitrarily much
$S \text{head} \sqsubseteq [1, \pi_1] \quad S \triangleq F(1, \exists) \alpha$	$S \text{head}$ is simple, which gives $R \circ (S^\circ S) \sqsubseteq Q$: between two paths S builds from one argument, equal cost and equal head already means Q

$$\text{PreservesRecip}(F) \wedge Q \in R \wedge 1 \in Q \wedge Q \subseteq Q \wedge R \circ R \subseteq R \wedge \text{MonoAlg}(F(\exists, 1)\alpha, Q) \wedge Rn(F(1, \exists)\alpha) \circ F(1, \exists)\alpha \in Q \quad (14.2c)$$

$$\Rightarrow (P(\frac{F(\exists, 1)}{\exists}) P(\frac{F(1, \exists)\alpha}{\exists} \text{est}(R))) \text{est}(R) \subseteq \frac{F(\exists, 1)\alpha}{\exists} \text{est}(R)$$

a least-cost path in a layered network, as a fold over the layers



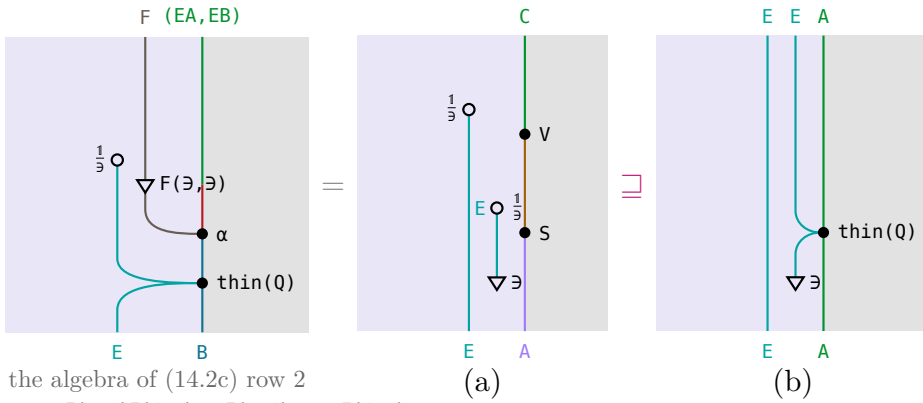
$\sqsubseteq$ (14.2d) under $(\sqsubseteq)$ monotonic: the whole chain runs inside the reduce, and the $\text{est}(R)$ behind it never moves.

$$Rn(F(1, \exists)\alpha) \circ F(1, \exists)\alpha \in Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists} \text{est}(R)) \subseteq \frac{F(\exists, \exists)\alpha}{\exists} \text{thin}(Q) \quad (14.2d)$$

thinning the algebra of a layered network costs no more than taking the program's two cases

$$\frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) P(\text{thin}(Q)) \cup \frac{F(\exists, \exists)\alpha}{\exists} \text{thin}(Q)$$

thinning each set the algebra's parts build, then taking the union, is below thinning the one set the whole algebra builds



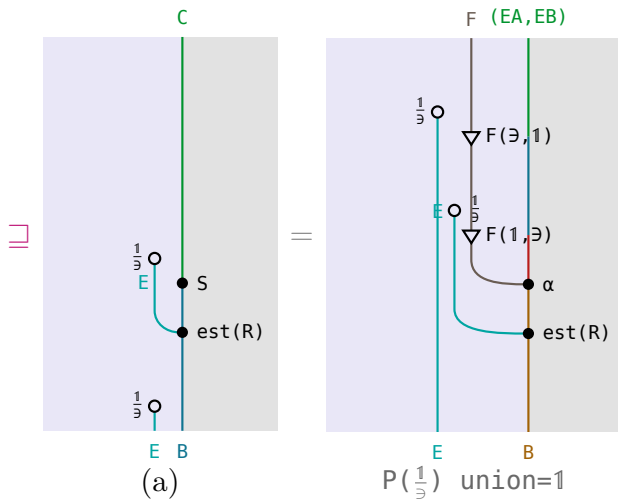
the algebra of (14.2c) row 2

(a) $\frac{F(\exists, 1)F(1, \exists)\alpha}{\exists} = \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \cup$

(b) $\cup \text{thin}(Q) \supseteq P(\text{thin}(Q)) \cup$ — (14.1b)

$$Rn(F(1, \exists)\alpha) \circ F(1, \exists)\alpha \in Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists} \text{est}(R)) \subseteq \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) P(\text{thin}(Q)) \cup$$

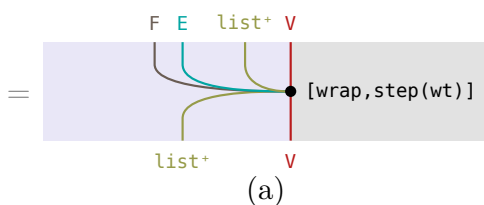
choosing an R-least member of each of those sets is below thinning each and taking the union



(a) $\frac{S}{\exists} \text{thin}(Q) \supseteq \frac{S}{\exists} \text{est}(R) \frac{1}{\exists}$ — (14.1b), (14.2b)

$$\text{cp}(F, \text{list}^+(V)) P(\alpha) \text{est}(\text{path}(\text{wt})) = [\text{wrap}, \text{step}(\text{wt})]$$

at the layered network, choosing an R-least path from each case is the program's two cases $[\text{wrap}, \text{step}]$



(a) α a map; $\frac{F(1, \exists)}{\exists} P(\alpha) \text{est}(R) = [\text{wrap}, \text{step}]$ — $\triangle$ cost

$$= \frac{F(\exists, 1)}{\exists} = 1 + \text{cpl} \triangle \text{cost}$$

§14.3. Implementing thin

(14.3a)

$\text{setify} : [A] \rightarrow EA$, $\text{cup} : EA \times EA \rightarrow EA$, $\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$, $\text{listcp} : F(L) \rightarrow LF$, $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P$ for P a connected preorder.

$\text{thinlist}(Q)$ is any $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ with $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q)$; one is $\langle [\text{nil}, \text{bump } Q] \rangle$, $\text{bump } Q (a, []) = [a]$, $\text{bump } Q (a, [b] \# xs) = (b \text{ } Q \text{ } a \rightarrow [a] \# xs, a \text{ } Q \text{ } b \rightarrow [b] \# xs, [a] \# [b] \# xs)$.

Binary thinning data: $S = (f_1 p_1) \cup (f_2 p_2)$ with p_1, p_2 coreflexive; Q a preorder with $Q \sqsubseteq R$ and both $f_1 p_1, f_2 p_2$ monotonic on Q ; P a connected preorder with both f_1, f_2 monotonic on P ; $g_i \triangleq \text{list}(f_i) \text{ filter}(p_i)$.

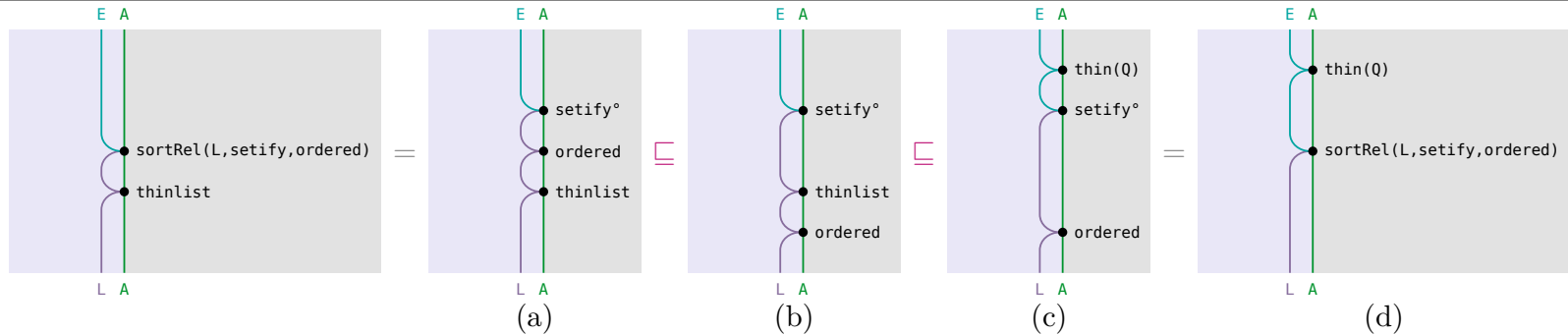
(14.3b)

the law	what it says
$\text{thinlist}(Q) \text{ } xs = [\text{minlist}(Q) \text{ } xs]$ (8.5), Q connected, xs non-empty	what thinning should come to when it can: one element
$\text{sort}(P) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(P)$ (8.6) — (14.3c)	thinning a sorted list is a thinning of the set — this is what $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ buys
$\text{sort}(P) \text{ minlist}(Q) \sqsubseteq \text{est}(Q)$ (8.7)	a minimum of the sorted list is a minimum of the set
$\text{sort}(f P f^\circ) \text{ list}(f) \sqsubseteq P(f) \text{ sort}(P)$ (8.8)	shunt a function through a sort
$\text{sort}(P) \text{ filter}(p) \sqsubseteq E(p) \text{ sort}(P)$ (8.9), p coreflexive	filtering a sorted list sorts the restricted set
$(\text{sort}(P) \times \text{sort}(P)) \text{ merge}(P) \sqsubseteq \text{cup sort}(P)$ (8.10)	merging two sorted lists sorts their union
$F(\text{sort}(P)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(FP)$ (8.11), F linear	listcp is the list implementation of the cartesian product $\text{cp}(F)$

$\text{Map}(\text{setify}) \wedge \text{Coreflexive}(\text{ordered}) \wedge \text{thinlist} \sqsubseteq \text{subseq} \wedge \text{ordered subseq} \sqsubseteq \text{subseq ordered} \wedge$ (14.3c)

$\text{thinlist setify} \sqsubseteq \text{setify thin}(Q) \Rightarrow \text{sortRel}(L, \text{setify}, \text{ordered}) \text{ thinlist} \sqsubseteq \text{thin}(Q) \text{ sortRel}(L, \text{setify}, \text{ordered})$

a thinning of the sorted list lists a thinning of the set — P a connected preorder, $\text{thinlist}(Q) \sqsubseteq \text{subseq}$.



- (a) $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \text{ — } \triangleq \text{thinlist}$
 (b) $\text{ordered } P \text{ thinlist}(Q) \sqsubseteq \text{thinlist}(Q) \text{ ordered } P \text{ — } \triangleq \text{thinlist}$
 (c) $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q) \text{ — } \triangleq \text{thinlist}$, dom-laws, triple-chains
 (d) $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \text{ — } \triangleq \text{thinlist}$

$$\text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f, P) \wedge \text{sortF}(fP f^\circ) \text{listf} \sqsubseteq P(f) \text{sort}(P) \wedge \quad (14.3d)$$

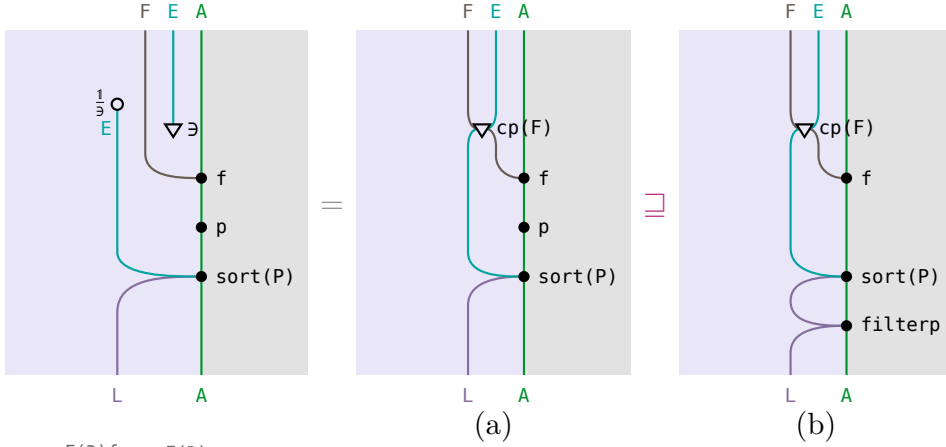
$$\text{sort}(P) \text{filterp} \sqsubseteq E(p) \text{sort}(P) \wedge F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}(F, A) \text{sortF}(F(P)) \Rightarrow F(\text{sort}(P)) \text{listcp} \text{listf}$$

$$\text{filterp} \sqsubseteq \frac{F(\exists)fp}{\exists} \text{sort}(P)$$

one sorted list built from sorted arguments, instead of a set built and then sorted — $f : FA \rightarrow A$ monotonic on P , p coreflexive, F linear.

$$\text{Map}(f) \wedge \text{sort}(P) \text{filterp} \sqsubseteq E(p) \text{sort}(P) \Rightarrow \text{cp}(F, A) P(f) \text{sort}(P) \text{filterp} \sqsubseteq \frac{F(\exists)fp}{\exists} \text{sort}(P)$$

sorting the f -images of the product and filtering by p is below sorting the set $F(\exists)fp$ builds

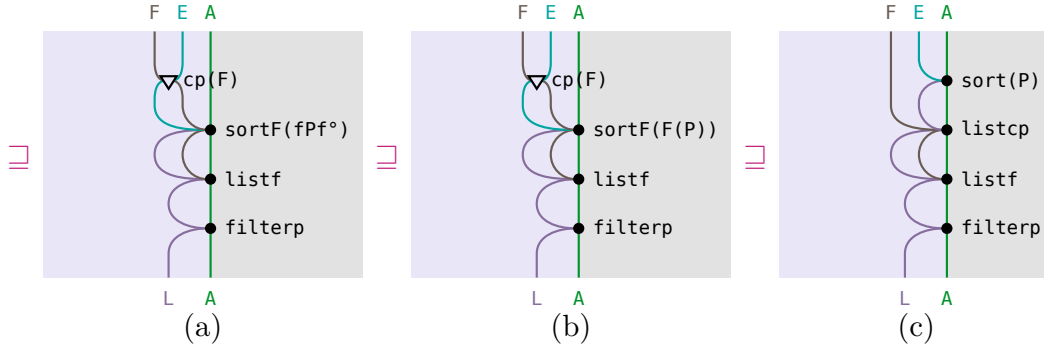


- (a) $\frac{F(\exists)fp}{\exists} = \frac{F(\exists)}{\exists} E(fp)$ — pow-laws, $\triangleq$ thinlist
 $E(f) = P(f)$, f a map — powrel-laws
 (b) $\text{sort}(P) \text{filterp} \sqsubseteq E(p) \text{sort}(P)$ — (14.3b)

$$\text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f, P) \wedge \text{sortF}(fP f^\circ) \text{listf} \sqsubseteq P(f) \text{sort}(P) \wedge$$

$$F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}(F, A) \text{sortF}(F(P)) \Rightarrow F(\text{sort}(P)) \text{listcp} \text{listf} \text{filterp} \sqsubseteq \text{cp}(F, A) P(f) \text{sort}(P) \text{filterp}$$

combining the arguments' sorted lists by listcp , mapping f and filtering is below sorting the f -images and filtering



- (a) $\text{sort}(fP f^\circ) \text{list}(f) \sqsubseteq P(f) \text{sort}(P)$ — (14.3b)
 (b) $FP \sqsubseteq fP f^\circ$ — mon-str, $\triangleq$ thinlist
 (c) $F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sort}(FP)$ — (14.3b)

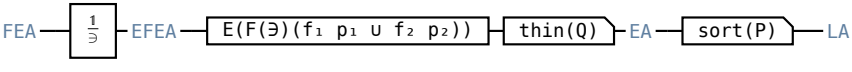
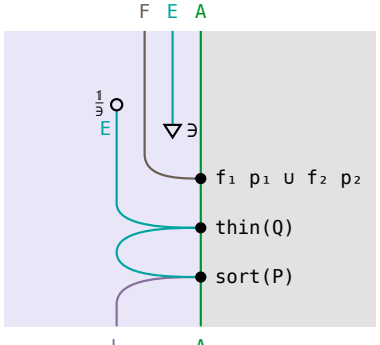
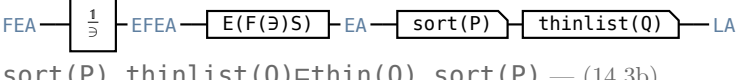
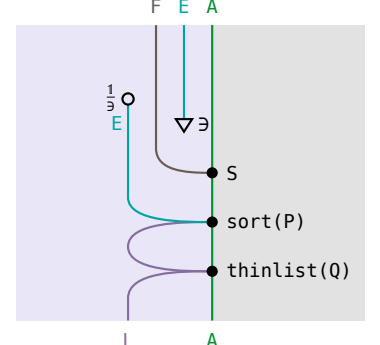
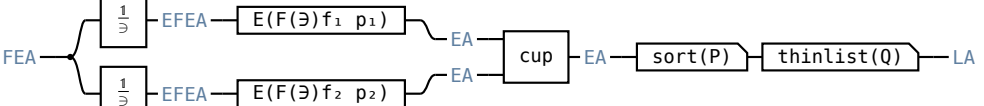
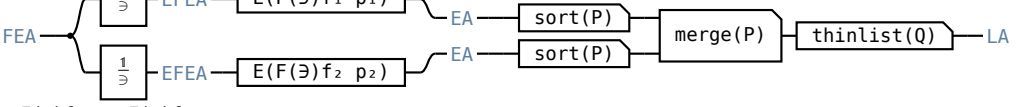
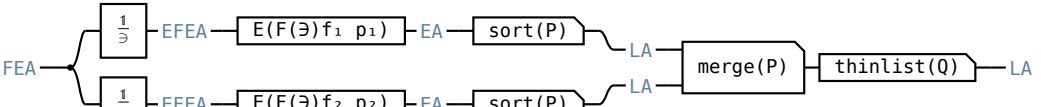
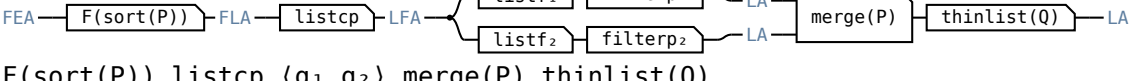
$$\begin{aligned}
 & \text{PreservesRecip}(F) \wedge \text{Map}(f_1) \wedge \text{Map}(f_2) \wedge Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R^\circ R^\circ \sqsubseteq R^\circ \wedge \text{MonoAlg}(f_1, p_1, Q) \wedge \\
 & \text{MonoAlg}(f_2, p_2, Q) \wedge (\forall X. \forall Y. X \sqsubseteq Y \implies \text{sort}F(X) \sqsubseteq \text{sort}F(Y)) \wedge \text{MonoAlg}(f_1, P) \wedge \text{MonoAlg}(f_2, P) \wedge \\
 & \text{sort}F(f_1 \text{ P } f_1^\circ) \text{list}f_1 \sqsubseteq P(f_1) \text{sort}(P) \wedge \text{sort}F(f_2 \text{ P } f_2^\circ) \text{list}f_2 \sqsubseteq P(f_2) \text{sort}(P) \wedge \\
 & \text{sort}(P) \text{filter}p_1 \sqsubseteq E(p_1) \text{sort}(P) \wedge \text{sort}(P) \text{filter}p_2 \sqsubseteq E(p_2) \text{sort}(P) \wedge \\
 & F(\text{sort}(P)) \text{list}cp \sqsubseteq cp(F, A) \text{sort}F(F(P)) \wedge (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \sqsubseteq \text{cup} \text{sort}(P) \wedge \\
 & \text{sort}(P) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(P) \wedge \text{sort}(P) \text{minlist}(R) \sqsubseteq \text{est}(R) \wedge g_1 = \text{list}f_1 \text{ filter}p_1 \wedge \\
 & g_2 = \text{list}f_2 \text{ filter}p_2 \wedge S = f_1 \text{ p}_1 \cup f_2 \text{ p}_2 \implies (\text{list}cp(g_1, g_2) \text{merge}(P) \text{thinlist}(Q)) \text{minlist}(R) \sqsubseteq \frac{(S)}{3} \\
 & \text{est}(R)
 \end{aligned} \tag{14.3e}$$

a fold on sorted lists of partial solutions, thinned at every step — at $\triangleq \text{thinlist}$'s binary thinning data.

circuit	Hinze–Marsden
$\sqsubseteq$ (14.1.2g) at $f_1 p_1$ and $f_2 p_2$ monotonic on $Q \triangleq \text{thinlist}$	
$\sqsubseteq$ $\text{sort}(P) \text{ minlist}(R) \sqsubseteq \text{est}(R)$ — (14.3b) at its $Q \triangleq R$	
$\sqsubseteq$ fusion at (14.3f)	

$$\begin{aligned}
& \text{Map}(f_1) \wedge \text{Map}(f_2) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, P) \wedge \text{MonoAlg}(f_2, P) \wedge \quad (14.3f) \\
& \text{sortF}(f_1 \text{ Pf}_1^\circ) \text{listf}_1 \sqsubseteq P(f_1) \text{sort}(P) \wedge \text{sortF}(f_2 \text{ Pf}_2^\circ) \text{listf}_2 \sqsubseteq P(f_2) \text{sort}(P) \wedge \\
& \text{sort}(P) \text{filterp}_1 \sqsubseteq E(p_1) \text{sort}(P) \wedge \text{sort}(P) \text{filterp}_2 \sqsubseteq E(p_2) \text{sort}(P) \wedge \\
& F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}(F, A) \text{sortF}(F(P)) \wedge (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \sqsubseteq \text{cup} \text{sort}(P) \wedge \\
& \text{sort}(P) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(P) \Rightarrow F(\text{sort}(P)) \text{listcp}(\text{listf}_1 \text{ filterp}_1, \text{listf}_2 \text{ filterp}_2) \\
& \text{merge}(P) \text{thinlist}(Q) \sqsubseteq \frac{F(\exists)(f_1 \text{ p}_1 \cup f_2 \text{ p}_2)}{\exists} \text{thin}(Q) \text{sort}(P)
\end{aligned}$$

sorting the candidate set is what turns the thinning algebra into an algebra on lists — the side condition of binary thinning theorem's last step.

circuit	Hinze–Marsden
	
$\sqsubseteq$  $\text{sort}(P) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(P) \text{ — (14.3b)}$	
$=$  $\{ \frac{F(\exists)f_1 p_1}{\exists}, \frac{F(\exists)f_2 p_2}{\exists} \} \text{cup} \text{sort}(P) \text{thinlist}(Q)$ $S = (f_1 p_1) \cup (f_2 p_2) \text{ — } \triangleq \text{thinlist, then } \triangleq \text{cup}$	
$\sqsubseteq$  $\{ \frac{F(\exists)f_1 p_1}{\exists}, \frac{F(\exists)f_2 p_2}{\exists} \} (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \text{thinlist}(Q)$ $(\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \sqsubseteq \text{cup} \text{sort}(P) \text{ — (14.3b)}$	
$=$  $\{ \frac{F(\exists)f_1 p_1}{\exists} \text{sort}(P), \frac{F(\exists)f_2 p_2}{\exists} \text{sort}(P) \} \text{merge}(P) \text{thinlist}(Q)$ $\langle X, Y \rangle (\text{sort}(P) \times \text{sort}(P)) = \langle X \text{sort}(P), Y \text{sort}(P) \rangle \text{ — bdm-prod-laws}$	
 $\sqsubseteq F(\text{sort}(P)) \text{listcp} \langle g_1, g_2 \rangle \text{merge}(P) \text{thinlist}(Q)$ $(14.3d) \text{ at } f_1, p_1 \text{ and at } f_2, p_2, \text{ then } X \langle g_1, g_2 \rangle \sqsubseteq \langle Xg_1, Xg_2 \rangle \text{ — bdm-prod-laws; } g_i \triangleq \text{list}(f_i) \text{filter}(p_i) \text{ — } \triangleq \text{thinlist}$	

§14.4. The knapsack problem

(14.4a)

$\text{vol}, \text{wt} : \text{Item} \rightarrow \text{Real}, \text{value} \triangleq \text{list}(\text{vol}) \text{ sum}, \text{weight} \triangleq \text{list}(\text{wt}) \text{ sum}.$

$\text{subseq} = ([\text{nil}, \text{cons}] \cup [\text{nil}, \pi_2]) \text{ , , within } w \text{ the coreflexive on } xs \text{ with weight } xs \leq w, 0 \leq w.$

$R \triangleq \text{value} \geq \text{value}^\circ, Q \triangleq R \cap (\text{weight} \leq \text{weight}^\circ), P \triangleq R, .$

$FA = 1 + \text{Item} \times A, \text{listcp} = \text{wrap} + \text{cpr}, g_1 \triangleq \text{list}([\text{nil}, \text{cons}]) \text{ filter}(\text{within } w) = [\text{list}(\text{nil}), h_1],$
 $g_2 \triangleq \text{list}([\text{nil}, \pi_2]) = [\text{list}(\text{nil}), h_2].$

$h_1 \triangleq \text{list}(\text{cons}) \text{ filter}(\text{within } w), h_2 \triangleq \text{list}(\pi_2).$

(14.4b)

the law	what it says
$(1 \times R) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) R$ FALSE	not monotonic on R: a selection of greater value need not still fit once one more item goes in
$(1 \times Q) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) Q$ $(1 \times Q) \pi_2 \sqsubseteq \pi_2 Q$,	both halves are monotonic on Q once ties in value are broken by weight

$$\begin{aligned}
 & 0 \leq w \wedge (\forall i. 0 \leq \text{wt}(i)) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \\
 & \text{sortF}([\text{nil}, \text{cons}]R[\text{nil}, \text{cons}]^\circ) \text{listf}_1 \sqsubseteq P([\text{nil}, \text{cons}]) \text{sortP} \wedge \\
 & \text{sortF}([\text{nil}, \pi_2]R[\text{nil}, \pi_2]^\circ) \text{listf}_2 \sqsubseteq P([\text{nil}, \pi_2]) \text{sortP} \wedge \text{sortP } \text{filterp}_1 \sqsubseteq E(\text{within}(w)) \text{sortP} \wedge \\
 & F(\text{sortP}) \text{listcp} \sqsubseteq \text{cp}(F, [\text{Item}]) \text{sortF}(F(R)) \wedge (\text{sortP} \times \text{sortP}) \text{mergeP} \sqsubseteq \text{cup } \text{sortP} \wedge \text{sortP} \\
 & \text{thinlist} \sqsubseteq \text{thin}(Q(\text{vol}, \text{wt})) \text{sortP} \wedge \text{sortP } \text{minlist} \sqsubseteq \text{est}(R) \Rightarrow (\text{listcp}(\text{listf}_1 \text{ filterp}_1, \text{listf}_2) \\
 & \text{mergeP } \text{thinlist}) \text{minlist} \sqsubseteq \frac{\text{subseq } \text{within}(w)}{\supseteq} \text{est}(R)
 \end{aligned}$$

the knapsack problem, as a fold that thins the packings kept at each item

(14.4c)

circuit	Hinze–Marsden
subseq (within w) $\supseteq$ est(R)	
$\supseteq$ fusion, weights non-negative.	
binary thinning theorem, at $P^A R$, F linear, Q from (14.4b)	
listcp=wrap+cpr, $g_i = [\text{list}(\text{nil}), h_i] \rightarrow \text{within}$; minlist(R) is head, packings coming out in descending value	

§14.5. The paragraph problem

(14.5a)

```

Line=list+ Word, Para=list+ Line, FA=Word+Word×A, listcp=wrap+cpr.
new(a,xs)=[[a]]#xs, glue(a,xs)=[[a]#head(xs)]#tail(xs), partition≐([wrap wrap,new u glue]) : list+
Word→Para.

width≐([length,(length×1) plus succ]), 0≤length a, fits w the coreflexive on a line x with width x≤w, ok w the
coreflexive on [x]#xs with width x≤w.

white w x=w-width x, collect≐list(sqr) sum, waste w≐init list(white w) collect.

R≐(waste w)≤(waste w)°, Q≐Rn(head head°), P≐T, .

g1≐list([wrap wrap,new]), g2≐list([wrap wrap,glue]) filter(ok w), start≐wrap wrap wrap, h1≐list(new),
h2≐list(glue) filter(ok w).

```

(14.5b)

the law	what it says
(1×R) glue⊆glue R FALSE	glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this
(1×Q) new⊆new Q (1×Q) (glue (ok w))⊆glue (ok w)Q cons monotonic on collect≤collect°.	both halves are monotonic on Q once ties in waste are broken by the first line
merge T=cat; P≐head prefix head° also serves	T needs no sorting at all, and prefix is a linear order on first lines of paragraphs of one input

$$\begin{aligned}
& (\forall a. 0 \leq \text{len}(a)) \wedge (\forall a. \text{len}(a) \leq w) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{wrap} \\
& \text{wrap}, \text{new}] \tau [\text{wrap} \text{ wrap}, \text{new}]^\circ) \text{listf}_1 \sqsubseteq \text{P}([\text{wrap} \text{ wrap}, \text{new}]) \text{sortP} \wedge \text{sortF}([\text{wrap} \text{ wrap}, \text{glue}] \tau [\text{wrap} \\
& \text{wrap}, \text{glue}]^\circ) \text{listf}_2 \sqsubseteq \text{P}([\text{wrap} \text{ wrap}, \text{glue}]) \text{sortP} \wedge \text{sortP} \text{filterp}_2 \sqsubseteq \text{E}(\text{ok}(w)) \text{sortP} \wedge \\
& \text{F}(\text{sortP}) \text{listcp} \sqsubseteq \text{cp}(\text{F}, \text{list}^+((\text{list}^+ \text{Word}))) \text{sortF}(\text{F}(\tau)) \wedge (\text{sortP} \times \text{sortP}) \text{mergeP} \sqsubseteq \text{cup} \text{sortP} \wedge \text{sortP} \\
& \text{thinlist} \sqsubseteq \text{thin}(\text{Q}(\text{len}, w)) \text{sortP} \wedge \text{sortP} \text{minlist} \sqsubseteq \text{est}(\text{R}) \Rightarrow (\text{listcp}(\text{listf}_1, \text{listf}_2 \text{filterp}_2) \\
& \text{mergeP} \text{thinlist}) \text{minlist} \sqsubseteq \frac{\text{partition fits}(w)}{\supset} \text{est}(\text{R})
\end{aligned} \tag{14.5c}$$

a paragraph laid out as a fold that thins the layouts kept at each word

circuit	Hinze–Marsden
$ \begin{aligned} & \text{list}^+(\text{Word}) \xrightarrow{\frac{1}{3}} \text{E}(\text{list}^+(\text{Word})) \xrightarrow{\text{E}(\text{partition fits}(w))} \text{E}(\text{list}^+(\text{list}^+(\text{Word}))) \xrightarrow{\text{est}(\text{R})} \text{list}^+(\text{list}^+(\text{Word})) \\ & \frac{\text{partition list}^+(\text{fits } w)}{\supset} \text{est}(\text{R}) \end{aligned} $	
$ \begin{aligned} & \text{list}^+(\text{Word}) \xrightarrow{\frac{1}{3}} \text{E}(\text{list}^+(\text{Word})) \xrightarrow{\text{E}([\text{wrap wrap}, \text{new}] \cup [\text{wrap wrap}, \text{glue}] \text{ok}(w))} \text{E}(\text{list}^+(\text{list}^+(\text{Word}))) \xrightarrow{\text{est}(\text{R})} \text{list}^+(\text{list}^+(\text{Word})) \\ & = \frac{([\text{wrap wrap}, \text{new}] \cup [\text{wrap wrap}, \text{glue}] \text{ok}(w))}{\supset} \text{est}(\text{R}) \end{aligned} $ fusion, every word fits on a line by itself.	
$ \begin{aligned} & \text{list}^+(\text{Word}) \xrightarrow{\frac{1}{3}} \text{E}(\text{list}^+(\text{Word})) \xrightarrow{\text{E}([\text{wrap wrap}, \text{new}] \cup [\text{wrap wrap}, \text{glue}] \text{ok}(w))} \text{E}(\text{list}^+(\text{list}^+(\text{Word}))) \xrightarrow{\text{est}(\text{R})} \text{list}^+(\text{list}^+(\text{Word})) \\ & = \frac{([\text{wrap wrap}, \text{new}] \cup ([\text{wrap wrap}, \text{glue}] \text{ok}(w)))}{\supset} \text{est}(\text{R}) \end{aligned} $ the algebra as $(f_1 p_1) \cup (f_2 p_2)$, $p_1 \triangleq 1$ — $\triangleq$ thinlist.	
$ \begin{aligned} & \text{listcp}(\text{g}_1, \text{g}_2) \text{cat thinlist}(\text{Q}) \\ & \sqsubseteq \text{minlist}(\text{R}) \end{aligned} $ binary thinning theorem, at $\text{P} \triangleq \text{T}$ with merge $\tau = \text{cat}$, Q from (14.5b)	
$ \begin{aligned} & ([\text{start}, \text{cpr}(\text{h}_1, \text{h}_2) \text{cat thinlist}(\text{Q})]) \\ & = \text{minlist}(\text{R}) \\ & \text{listcp} = \text{wrap} + \text{cpr}, \text{g}_i \text{ along the coproduct} \text{ — } \triangleq \text{partition} \end{aligned} $	

§14.6. Bitonic tours

$\text{FA} = (\text{City} \times \text{City}) + (\text{City} \times \text{A})$, the base functor of cons-lists of length at least two; $\text{listcp} = \text{wrap} + \text{cpr}$; $\text{tc} : \text{City} \times \text{City} \rightarrow \text{Real}$, neither positive nor symmetric.

$\text{start}(a, b) = ([a, b], [a, b])$, $\text{dropl}(a, ([b] \# xs, ys)) = ([a] \# xs, [a] \# ys)$, $\text{dropr}(a, (xs, [b] \# ys)) = ([a] \# xs, [a] \# ys)$, $\text{tour} \triangleq ([\text{start}, \text{dropl} \cup \text{dropr}])$.

$\text{cost}(xs, ys) = \text{outcost } xs + \text{incost } ys$, $\text{outcost}[a_0, \dots, a_n] = \text{tc}(a_0, a_1) + \dots + \text{tc}(a_{n-1}, a_n)$, $\text{incost}[a_0, \dots, a_n] = \text{tc}(a_1, a_0) + \dots + \text{tc}(a_n, a_{n-1})$.

$\text{next} \triangleq \text{tail head}$, $\text{next2} \triangleq \text{next} \times \text{next}$, $\text{head2} \triangleq \text{head} \times \text{head}$, $\text{R} \triangleq \text{cost} \leq \text{cost}^\circ$, $\text{Q} \triangleq \text{Rn}(\text{next2 next2}^\circ) \text{n}(\text{head2 head2}^\circ)$, $\text{P} \triangleq \text{T}$, $\text{g}_1 \triangleq \text{list}([\text{start}, \text{dropl}])$, $\text{g}_2 \triangleq \text{list}([\text{start}, \text{dropr}])$.

the law	what it says
$(1 \times \text{R}) \text{dropl} \sqsubseteq \text{dropl } \text{R}$ FALSE $(1 \times \text{R}) \text{dropr} \sqsubseteq \text{dropr } \text{R}$ FALSE	neither drop is monotonic on R: the two edges it adds and removes depend on head and next of both lists
$(1 \times \text{Q}) \text{dropl} \sqsubseteq \text{dropl } \text{Q}$ $(1 \times \text{Q}) \text{dropr} \sqsubseteq \text{dropr } \text{Q}$	both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input

$ \begin{aligned} & (\forall X. \forall Y. X \sqsubseteq Y \implies \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \\ & \text{sortF}([\text{start}, \text{droplFn}] \sqcup [\text{start}, \text{droplFn}]^\circ) g_1 \sqsubseteq P([\text{start}, \text{droplFn}]) \text{sortP} \wedge \\ & \text{sortF}([\text{start}, \text{droprFn}] \sqcup [\text{start}, \text{droprFn}]^\circ) g_2 \sqsubseteq P([\text{start}, \text{droprFn}]) \text{sortP} \wedge \\ & F(\text{sortP}) \text{listcp} \sqsubseteq \text{cp}(F, \text{dTour}(\text{City})) \text{sortF}(F(\tau)) \wedge (\text{sortP} \times \text{sortP}) \text{cat} \sqsubseteq \text{cup} \text{sortP} \wedge \text{sortP} \\ & \text{thinlist} \sqsubseteq \text{thin}(Q(\text{tc})) \text{sortP} \wedge \text{sortP} \text{minlist} \sqsubseteq \text{est}(R) \implies (\text{listcp}(g_1, g_2) \text{cat} \text{thinlist}) \text{minlist} \sqsubseteq \\ & \quad \underbrace{(\text{tourAlg})}_{\exists} \text{est}(R) \end{aligned} $ a least-cost bitonic tour, as a fold that thins the tours kept at each city	
circuit	Hinze–Marsden
$ \begin{aligned} & \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{\exists}} E(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E(\text{tourAlg})} E(\text{Journey}(\text{City})^2) \xrightarrow{\text{est}(R)} \text{Journey}(\text{City}) \\ & \frac{\text{tour}}{\exists} \text{est}(R) \end{aligned} $	
$ \begin{aligned} & \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{\exists}} E(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E([\text{start}, \text{droplFn}] \sqcup [\text{start}, \text{droprFn}])} E(\text{Journey}(\text{City})^2) \xrightarrow{\text{est}(R)} \text{Journey}(\text{City}) \\ & = \frac{([\text{start}, \text{dropl} \sqcup \text{dropr}])}{\exists} \text{est}(R) \\ & \text{tour} \triangleq ([\text{start}, \text{dropl} \sqcup \text{dropr}]) \triangleq \text{tour} \end{aligned} $	
$ \begin{aligned} & \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{\exists}} E(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E([\text{start}, \text{droplFn}] \sqcup [\text{start}, \text{droprFn}])} E(\text{Journey}(\text{City})^2) \xrightarrow{\text{est}(R)} \text{Journey}(\text{City}) \\ & \sqsubseteq (\text{listcp}(g_1, g_2) \text{cat} \text{thinlist}(Q)) \text{minlist}(R) \end{aligned} $ binary thinning theorem, at $P \triangleq T$ with merge $T = \text{cat}$, Q from (14.6b)	
$ \begin{aligned} & ([\text{start wrap}, \text{cpr}(\text{list}(\text{dropl}), \text{list}(\text{dropr})) \text{cat} \\ & = \text{thinlist}(Q)] \text{minlist}(R) \\ & \text{listcp} = \text{wrap} + \text{cpr}, g_i = [\text{list}(\text{start}), \text{list}(\text{drop}_i)] \triangleq \text{tour}; \\ & \text{quadratic, two tours added per step} \end{aligned} $	